

GET it digital

Modul 11: Electrical machines

Stand: 18. Februar 2026



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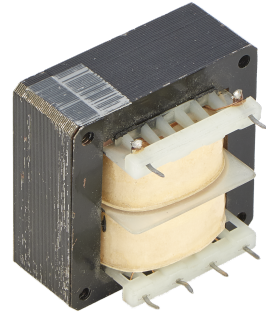
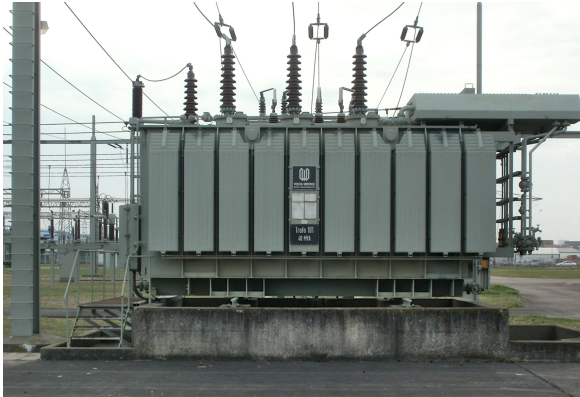
<https://getitdigital.uni-wuppertal.de/module/modul-11-elektrische-maschinen>

Learning objectives: Electrical machinery

The students

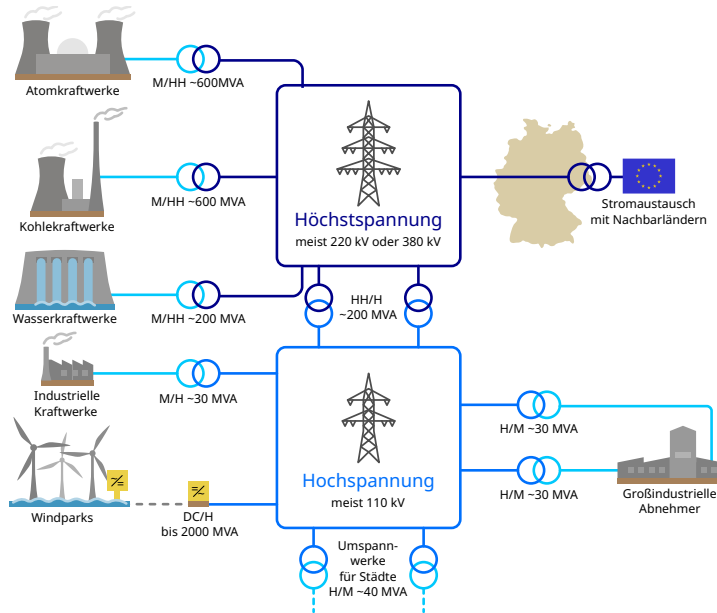
- ▶ are familiar with the various types of electrical machines.
- ▶ can solve basic problems in the field of electrical machines.

The transformer

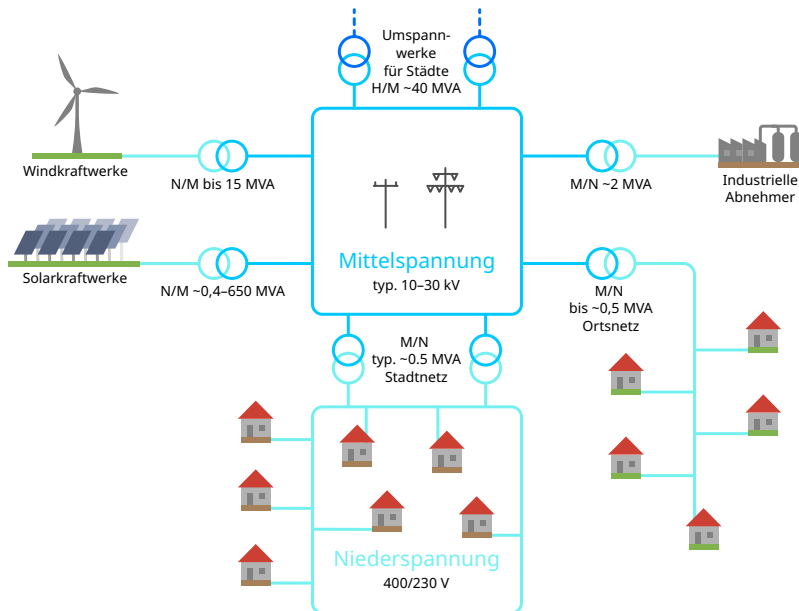


- ▶ Conversion of electrical energy $\rightarrow$ electrical energy
- ▶ Voltage converter

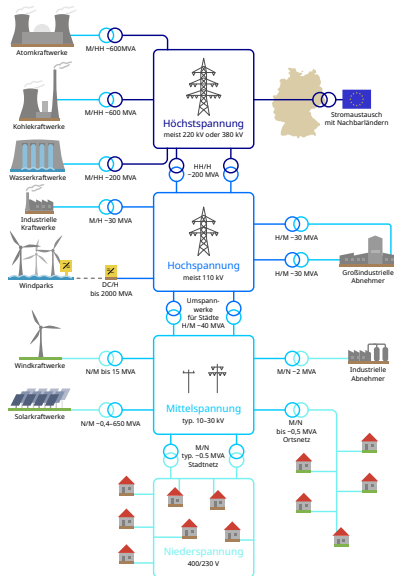
Public electricity supply



Public electricity supply



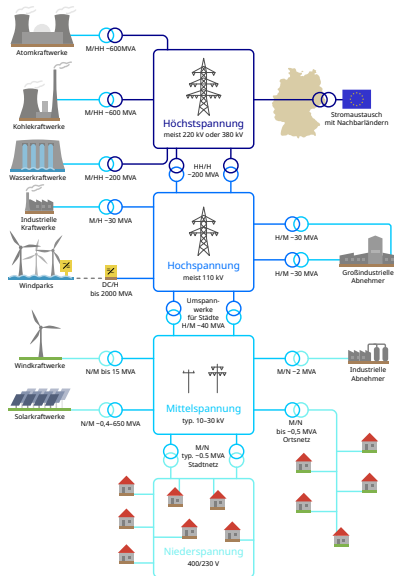
Public electricity supply



Losses proportional to the square of the current.

$$P_v = 3 \cdot R_L \cdot I_L^2$$

Public electricity supply



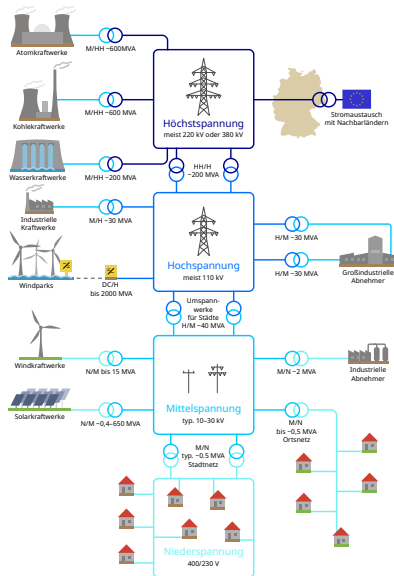
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Transferred power:

$$\underline{S}_N = \sqrt{3} \cdot \underline{U}_N \cdot \underline{I}_L^*$$

Public electricity supply



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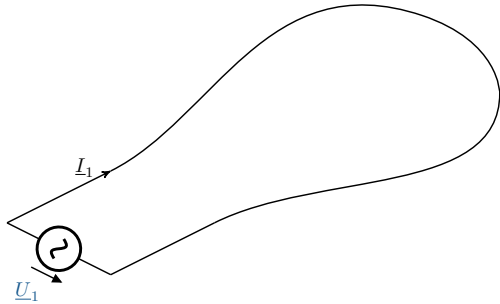
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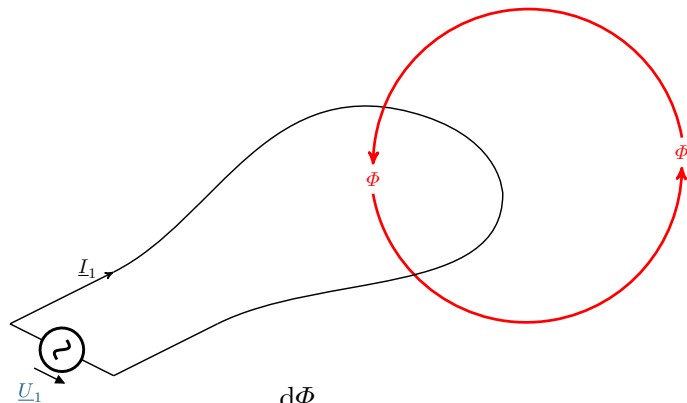
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Losses proportional to $\frac{1}{\underline{U}_N^2}$

Principle of the transformer



Principle of the transformer

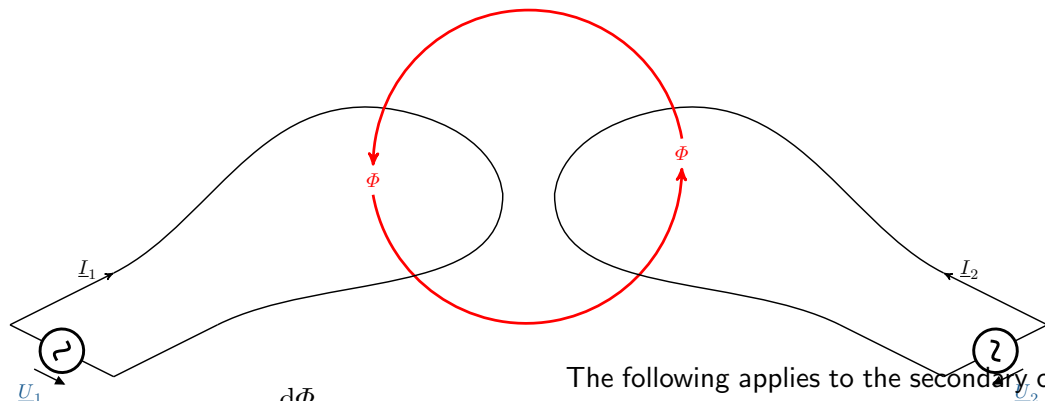


$$u_i = -N \cdot \frac{d\Phi}{dt}$$

$$u = \hat{U} \cdot \cos(\omega t)$$

$$\Phi = \frac{\sqrt{2} \cdot \underline{U}_1}{2\pi \cdot N_1 \cdot f}$$

Principle of the transformer

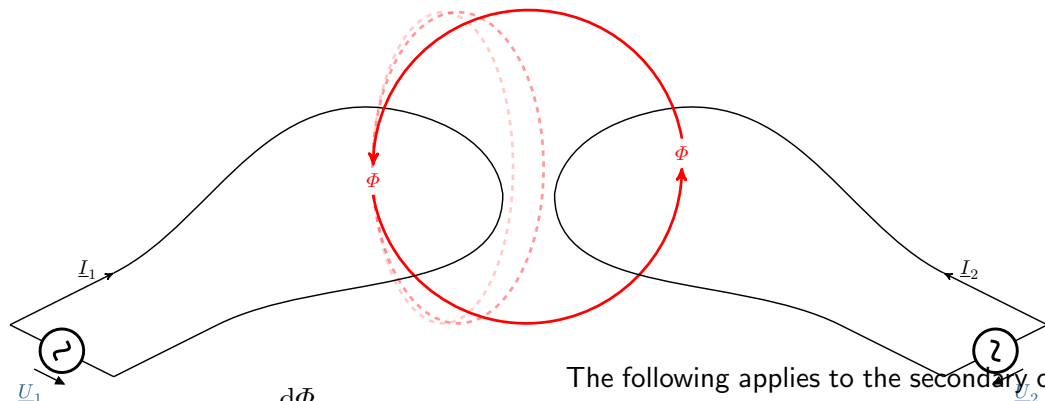


The following applies to the secondary coil:

$$\begin{aligned}u_i &= -N \cdot \frac{d\Phi}{dt} \\u &= \hat{U} \cdot \cos(\omega t) \\\Phi &= \frac{\sqrt{2} \cdot \underline{U}_1}{2\pi \cdot N_1 \cdot f}\end{aligned}$$

$$\begin{aligned}u_2 &= -N_2 \cdot \frac{d\Phi}{dt} \\\frac{\underline{U}_1}{\underline{U}_2} &= \frac{-N_1 \cdot \frac{d\Phi}{dt}}{-N_2 \cdot \frac{d\Phi}{dt}} = \frac{N_1}{N_2} = \ddot{u}\end{aligned}$$

Principle of the transformer



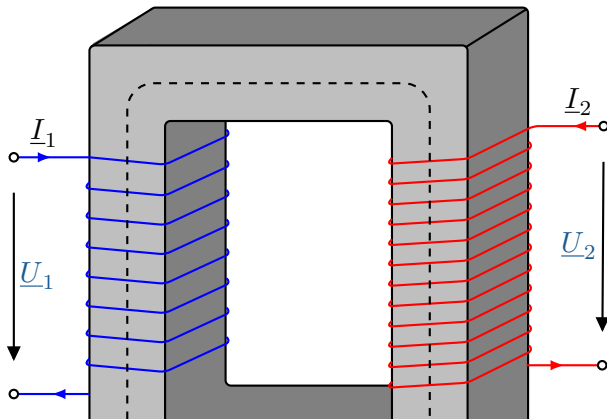
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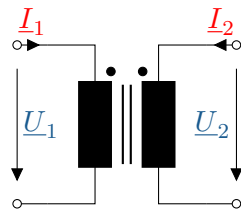
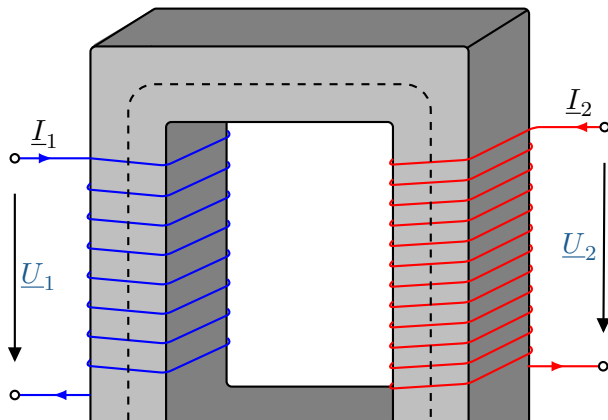
Transformer structure

In simple terms, a transformer consists of two coils that are coupled together by means of a magnetically conductive core (see Figure ??). The core is made of a ferromagnetic material. Due to its high permeability, the ferromagnetic material has the ability to conduct the magnetic flux between two coils as efficiently as possible (see Module 6). However, effects such as hysteresis and eddy current losses interfere with the resistance-free flow. These effects are referred to as iron losses.



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$\ddot{u} < 1 \rightarrow$ Voltage high

$\ddot{u} > 1 \rightarrow$ Voltage low

Example: Ideal three-phase mains transformer

An ideal three-phase power transformer has an apparent power of $\underline{S}_N = 100 \text{ kVA}$, an upper-side voltage of $\underline{U}_1 = 20 \text{ kV}$ (delta connection) and a lower-side voltage of $\underline{U}_2 = 400 \text{ V}$ (delta connection).

a) What is the gear ratio?

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a) What is the gear ratio?

$$\begin{aligned}\ddot{u} &= \frac{\underline{U}_1}{\underline{U}_2} = \frac{N_1}{N_2} \\ &= \frac{20 \text{ kV}}{400 \text{ V}} = 50\end{aligned}$$

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- b) What are the primary and secondary currents in nominal operation at a power factor of $\cos(\varphi) = 1$?

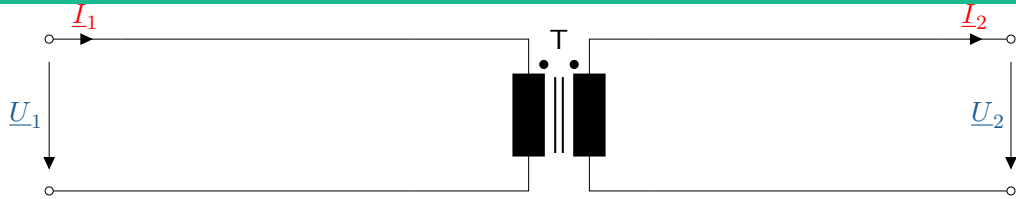
$$\cos(\varphi) = 1 \rightarrow |\underline{S}| = P$$

$$P = U \cdot I$$

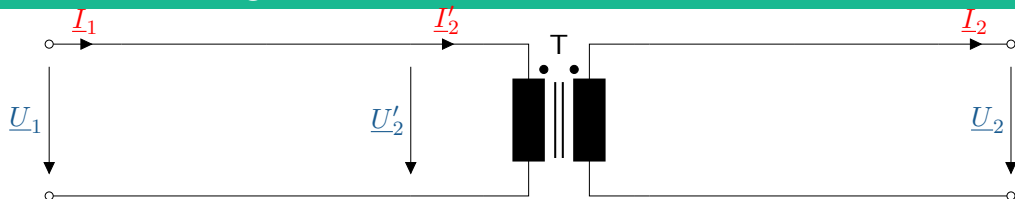
$$I_1 = \frac{P}{U} = \frac{100 \text{ kW}}{20 \text{ kV}} = 5 \text{ A}$$

$$I_2 = \frac{P}{U} = \frac{100 \text{ kW}}{400 \text{ V}} = 250 \text{ A}$$

Euivalent circuit diagram



Euivalent circuit diagram

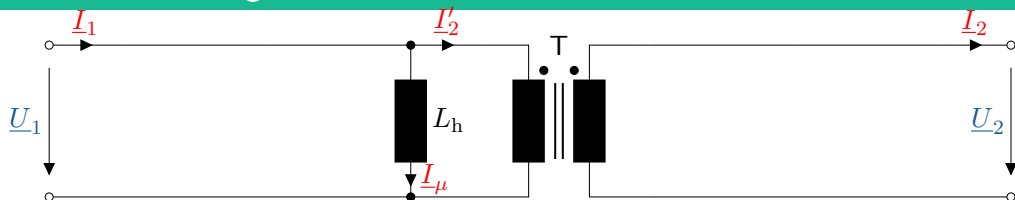


$$\underline{U}_1 = \underline{U}'_2 = \frac{N_1}{N_2} \cdot \underline{U}_2$$

$$\underline{I}_1 = \underline{I}'_2 = \frac{N_2}{N_1} \cdot \underline{I}_2$$

To convert the voltage to the primary side, the voltage $\underline{U}'_2$ is multiplied by the transformation ratio. To calculate the incoming current $\underline{I}_1$, the current $\underline{I}_2$ is multiplied by the inverse transformation ratio and then added to the magnetising current $\underline{I}_\mu$ (Equation ??).

Euivalent circuit diagram

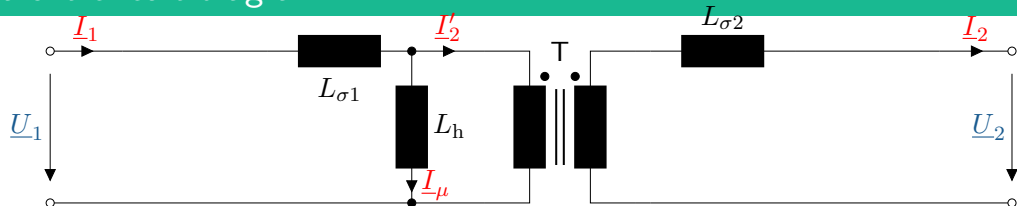


$$\underline{U}_1 = \underline{U}'_2 = \frac{N_1}{N_2} \cdot \underline{U}_2$$

$$\underline{I}_1 = \underline{I}'_2 + \underline{I}_\mu = \frac{N_2}{N_1} \cdot \underline{I}_2 + \underline{I}_\mu$$

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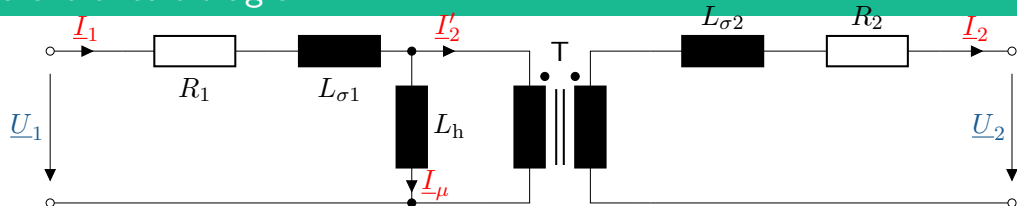


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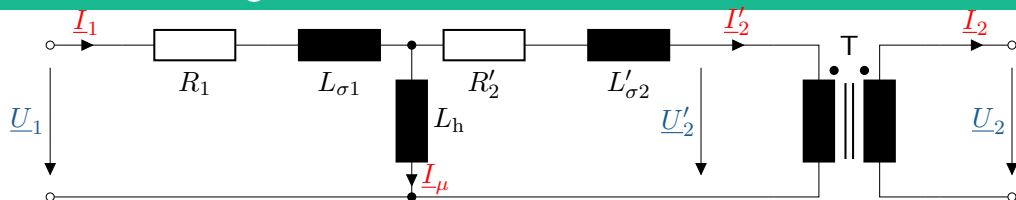


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$$L'_{\sigma 2} = \left(\frac{N_1}{N_2} \right)^2 \cdot L_{\sigma 2}$$

$$R'_2 = \left(\frac{N_1}{N_2} \right)^2 \cdot R_2$$

The ideal transformer on the right is

Example: Real three-phase mains transformer

A real three-phase mains transformer ($f = 50 \text{ Hz}$) has the following data:

- ▶ Power: $\underline{S}_N = 100 \text{ kVA}$
- ▶ Overvoltage $\underline{U}_1 = 20 \text{ kV}$
- ▶ Transmission ratio: 50
- ▶ $L_h = 500 \text{ H}$
- ▶ $L_{\sigma 1} = L'_{\sigma 2} = 190 \text{ mH}$
- ▶ $R_1 = R'_2 = 30 \Omega$

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a) What is the open-circuit voltage $\underline{U}_2$ for this transmission ratio?

$$\begin{aligned}\underline{U}'_2 &= \underline{U}_1 \cdot \frac{\underline{Z}_h}{\underline{Z}_h + \underline{Z}_{\sigma 1} + R_1} \\ &= 19,992 \text{ kV} \cdot e^{j0,011^\circ} \\ \underline{U}_2 &= 399,85 \text{ V} \cdot e^{j0,011^\circ}\end{aligned}$$

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$$\begin{aligned}\underline{I}_1 &= \frac{\underline{U}_1}{\underline{Z}_h + \underline{Z}_{\sigma 1} + R_1} \\ &= \frac{20 \text{ kV}}{157,139 \text{ k}\Omega \cdot e^{j89,989^\circ}} \\ &= 127,28 \text{ mA} \cdot e^{-j89,989^\circ}\end{aligned}$$

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$$\begin{aligned}\underline{S}_v &= \underline{U} \cdot \underline{I}^* \\ &= 20 \text{ kV} \cdot (127,28 \text{ mA} \cdot e^{-j89,989^\circ})^* \\ &= 2,546 \text{ kVA} \cdot e^{j89,989^\circ}\end{aligned}$$

Advantages:

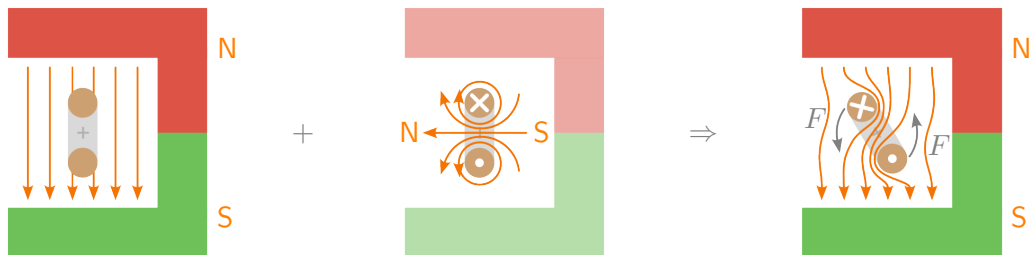
- ▶ Simpler and more cost-effective
Design of the power converter
- ▶ High control dynamics
- ▶ Direct operation of the machine
with
accumulators possible
- ▶ High overload capacity

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Disadvantages:

- ▶ High construction costs
- ▶ High maintenance (brushes)
- ▶ Low power density
- ▶ Expenses



$$F = B \cdot I \cdot \ell \cdot N$$

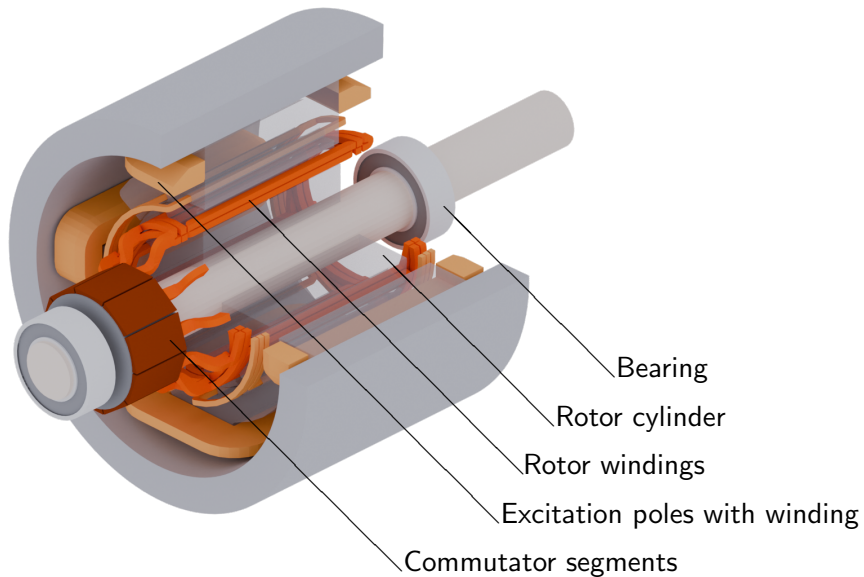
Example: Power of a direct current machine

A direct current motor has a magnetic flux density of $B = 0.8 \text{ T}$ in the air gap. There are a total of $N = 400$ armature wires under the poles, through which a current of $I = 10 \text{ A}$ flows. The effective conductor length is $\ell = 150 \text{ mm}$. Calculate the force F at the circumference of the anchor.

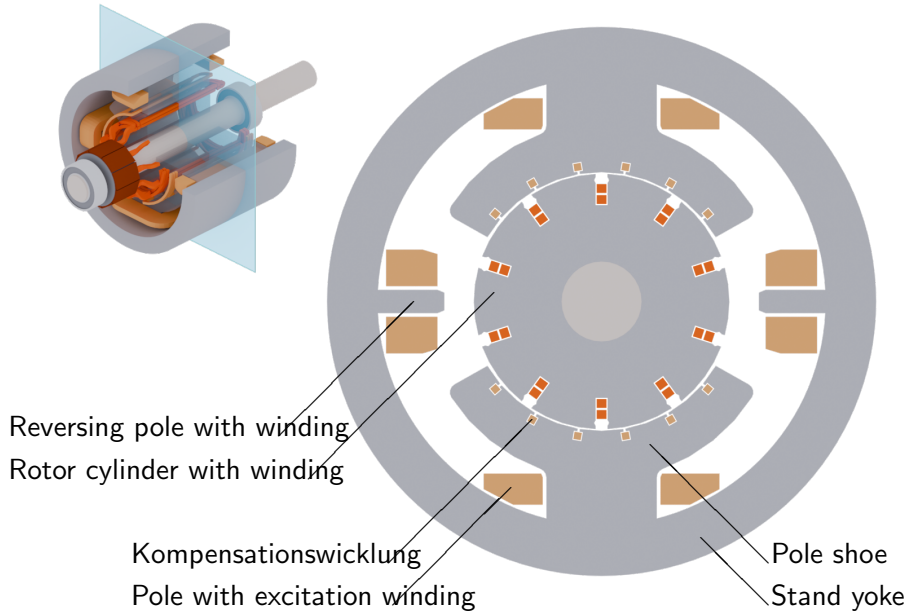
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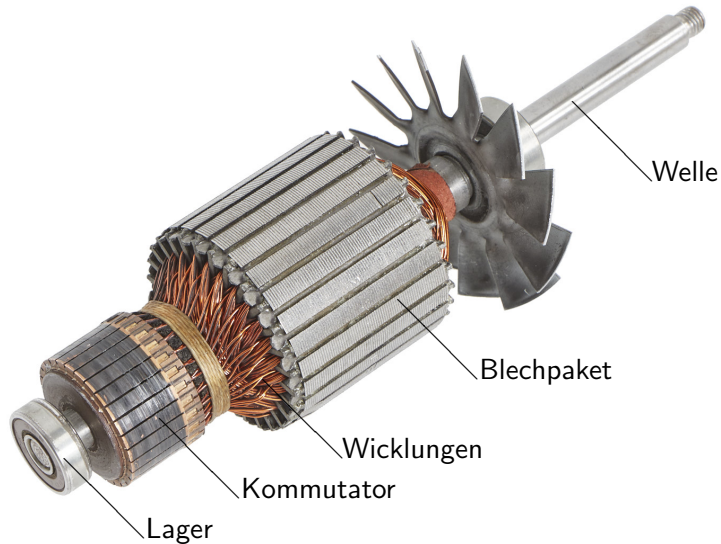
$$\begin{aligned} F &= B \cdot I \cdot \ell \cdot N \\ &= 0,8 \frac{\text{Vs}}{\text{m}^2} \cdot 10 \text{ A} \cdot 0,15 \text{ m} \cdot 400 \\ &= 480 \frac{\text{kg} \cdot \text{m}^2 \cdot \text{s} \cdot \text{A} \cdot \text{m}}{\text{s}^3 \cdot \text{A} \cdot \text{m}^2} = 480 \text{ N} \end{aligned}$$

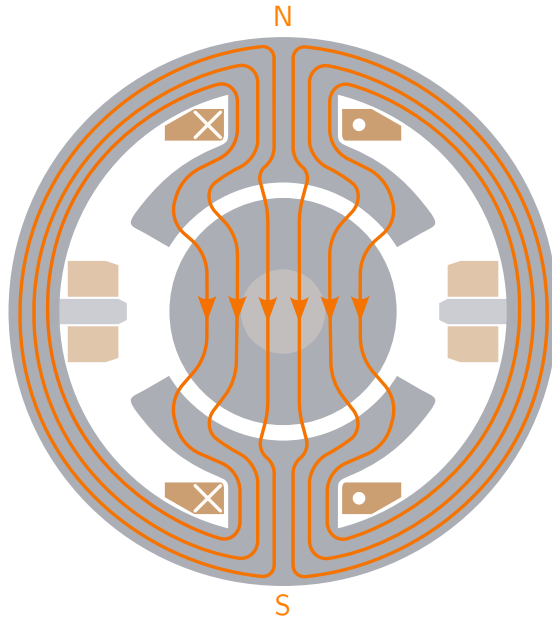


Structure



Rotor and commutator





$$U_q = K \cdot \Phi \cdot \omega$$

- ▶ Induced voltage (source voltage): U_q [V]
- ▶ Armature constant: K [1]
- ▶ Flux: Φ [Vs]
- ▶ Angular velocity: ω_{mech} [rad/s]

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$$P_i = U_q \cdot I_A = P_{\text{mech}} = M_i \cdot \omega$$

- ▶ internal electrical power: P_i [W]

$$M_i = K \cdot \Phi \cdot I_A$$

- ▶ Internal electric moment: M_i [Nm]
- ▶ Anchor constant: K [1]
- ▶ Flux: Φ [Vs]
- ▶ Anchor current: I_A [A]

$$M_i = K \cdot \Phi \cdot I_A$$

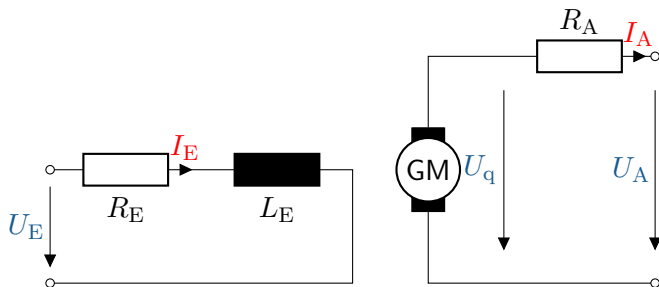
- ▶ Internal electric moment: M_i [Nm]
- ▶ Anchor constant: K [1]
- ▶ Flux: Φ [Vs]
- ▶ Anchor current: I_A [A]

$$M = M_i - M_V$$

$$M = \eta_1 \cdot M_i$$

- ▶ Loss moment: M_V [Nm]
- ▶ Efficiency: η [1]

Externally excited direct current machine



$$U_A = U_q + I_A \cdot R_A$$

$$n = \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi (K \cdot \Phi)^2}$$

$$I_A = \frac{M_i}{K \cdot \Phi}$$

Example: Separately excited DC machine

A separately excited DC machine with the armature constant $K = \frac{1}{2\pi}$ has a no-load speed of $n = 1200 \frac{1}{\text{min}}$ at an armature voltage of $U_A = 400 \text{ V}$. The armature resistance is $R_A = 2.3 \Omega$.

a) How large is the pathogen flow? Φ ?

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a) How large is the pathogen flow? Φ ?

When idling, the internal torque M_i is zero. The load-dependent speed drop therefore does not apply:

$$n = \frac{U_A}{2\pi K \cdot \Phi}$$
$$\Phi = \frac{U_A}{2\pi K \cdot n} = \frac{400 \text{ V}}{\frac{2\pi}{2\pi} \cdot 1200 \frac{1}{\text{min}} \cdot 60 \frac{1}{\text{s}}} = 20 \text{ Vs}$$

b) How fast does the machine rotate at an internal torque of $M_i = 10 \text{ Nm}$?

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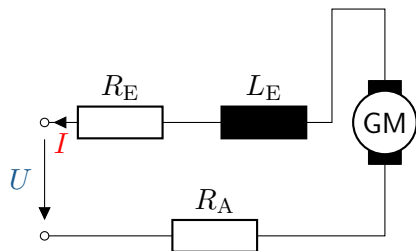
$$\begin{aligned} n &= \frac{U_A}{2\pi K \cdot \Phi} - \frac{R_A \cdot M_i}{2\pi (K \cdot \Phi)^2} \\ &= \frac{400 \text{ V}}{\frac{2\pi}{2\pi} \cdot 20 \text{ Vs}} - \frac{2,3 \Omega \cdot 10 \text{ Nm}}{2\pi \left(\frac{1}{2\pi} \cdot 20 \text{ Vs}\right)^2} \\ &= 20 \frac{1}{\text{s}} - 0,362 \frac{1}{\text{s}} = 19,638 \frac{1}{\text{s}} = 1178,3 \frac{1}{\text{min}} \end{aligned}$$

b) How fast does the machine rotate at an internal torque of $M_i = 10 \text{ Nm}$?

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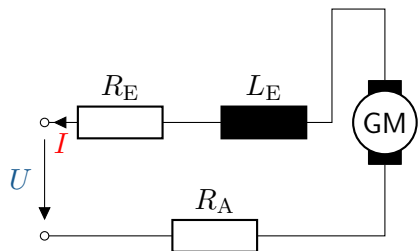
c) How big is the anchor current?

$$I_A = \frac{M_i}{K \cdot \Phi} = \frac{10 \text{ Nm}}{\frac{1}{2\pi} \cdot 20 \text{ Vs}} = 3,61 \text{ A}$$

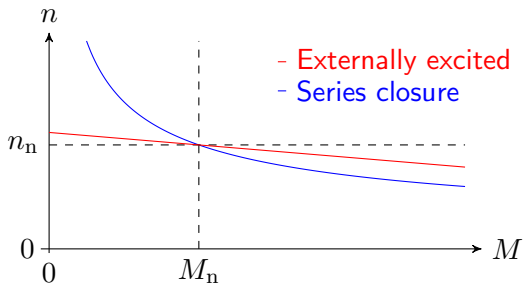


- ▶ Can also be used for alternating voltage
- ▶ high tightening torque

Series closure machine



- ▶ Can also be used for alternating voltage
- ▶ high tightening torque



The synchronous machine rotates at a constant speed in synchronisation with the mains. **Advantages:**

- ▶ Robust and maintenance-free.
- ▶ High control dynamics.
- ▶ High power density with low construction volume.
- ▶ Overload capacity.

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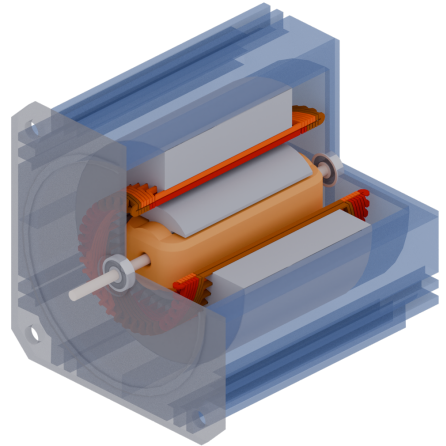
- ▶ Robust and maintenance-free.
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- ▶ Overload capacity.

Disadvantages:

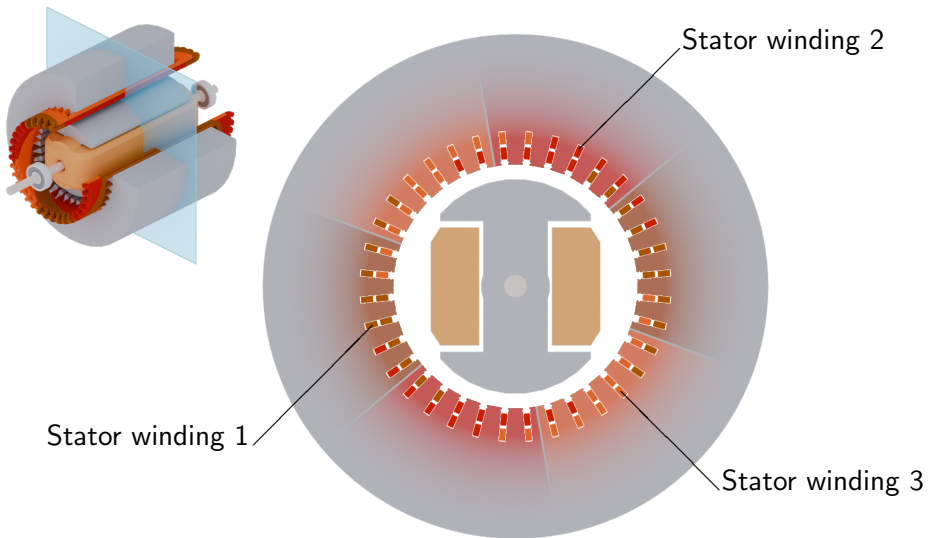
- ▶ Complex converter technology.
- ▶ Complex control system.
- ▶ Controlled operation only possible with great effort or not possible at all.

Structure of a separately excited synchronous machine

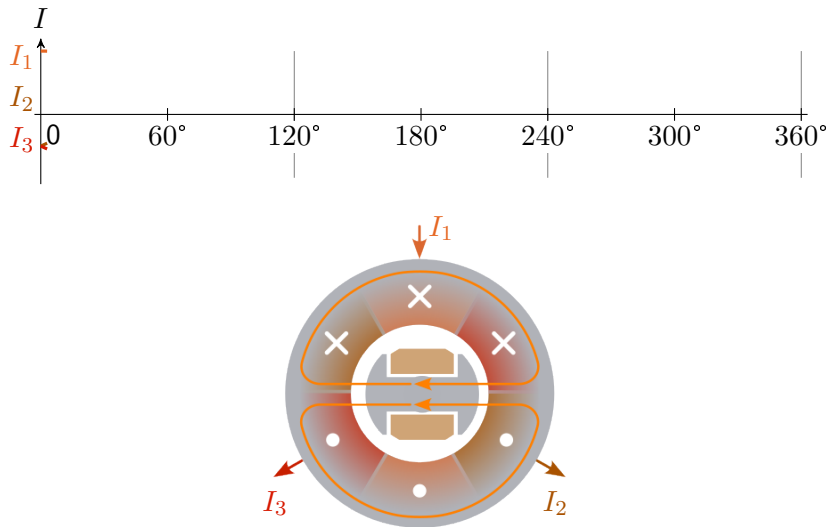
- ▶ Three stator windings offset by 120° from each other.
- ▶ Laminated stator design, as it is subjected to an alternating magnetic field.
- ▶ The exciter winding is located in the rotor.
- ▶ The rotor is partially unlaminated.



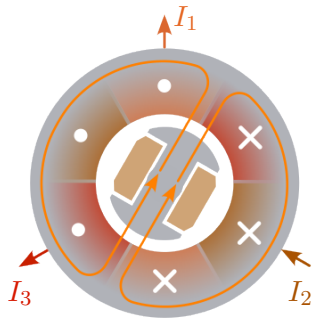
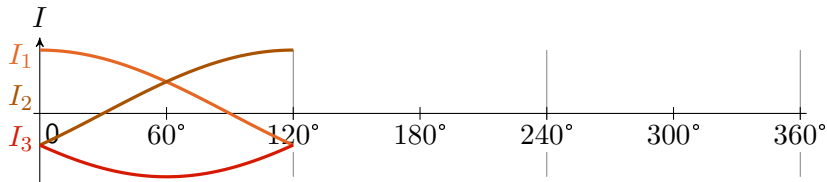
Structure of a separately excited synchronous machine



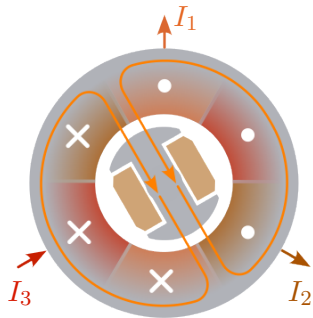
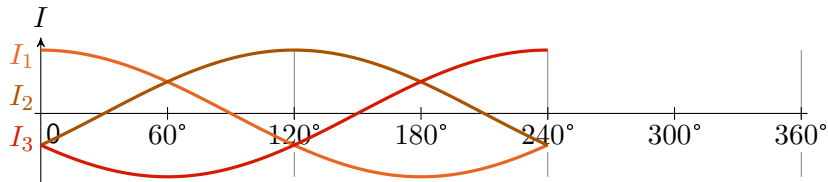
Field gradients



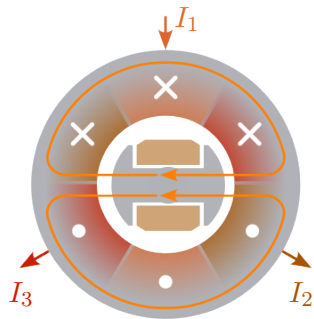
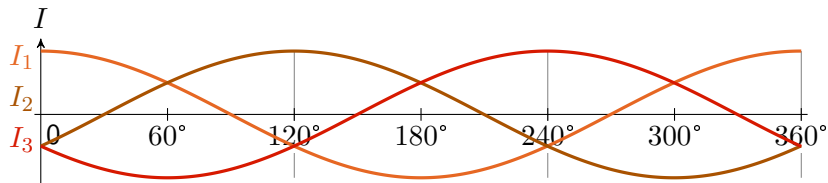
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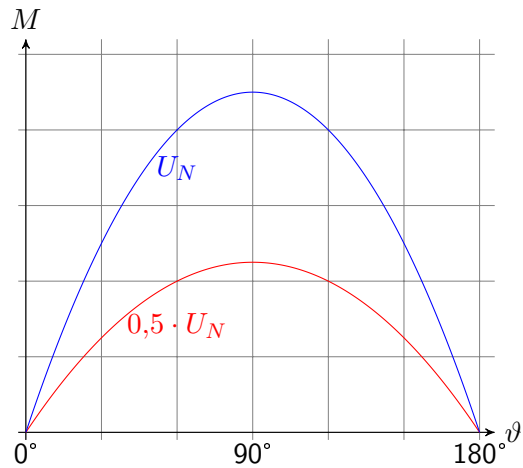


Field gradients

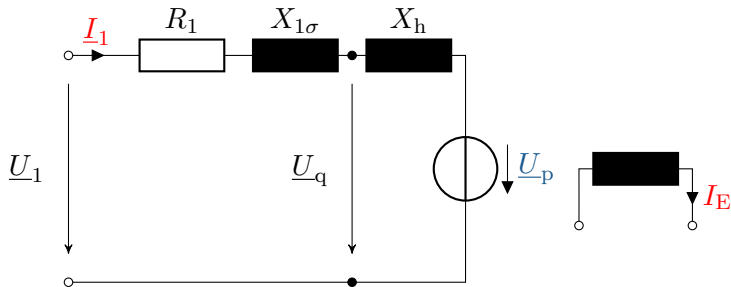


- ▶ The stator flux is created by the superposition of the fluxes of the three windings.

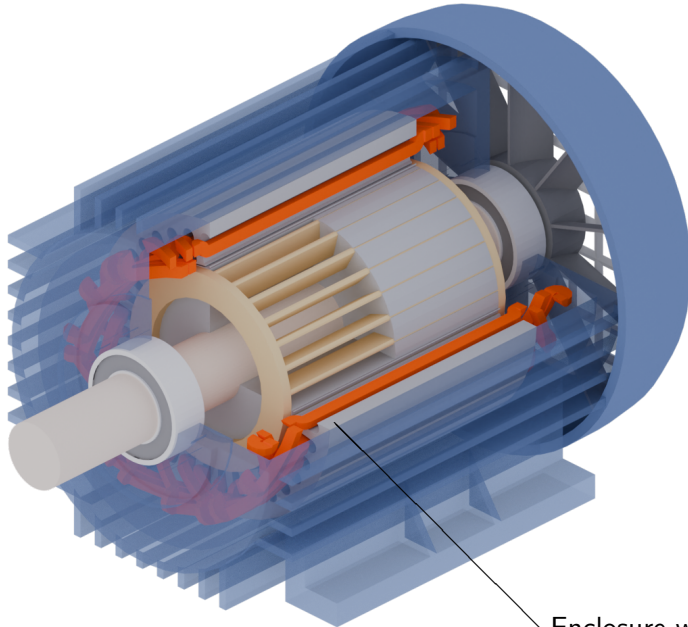
- ▶ The stator flux is created by the superposition of the fluxes of the three windings.
- ▶ The current flow is generated by the excitation winding.



Complex single-phase ESB of the full-pole generator (e.g. in grid operation):

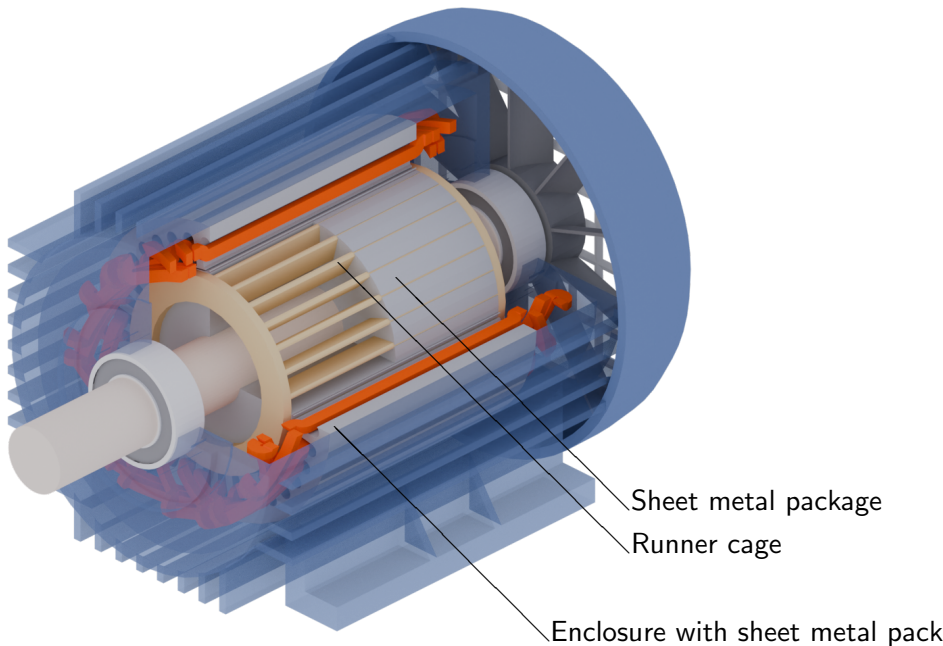


Asynchronous machine design

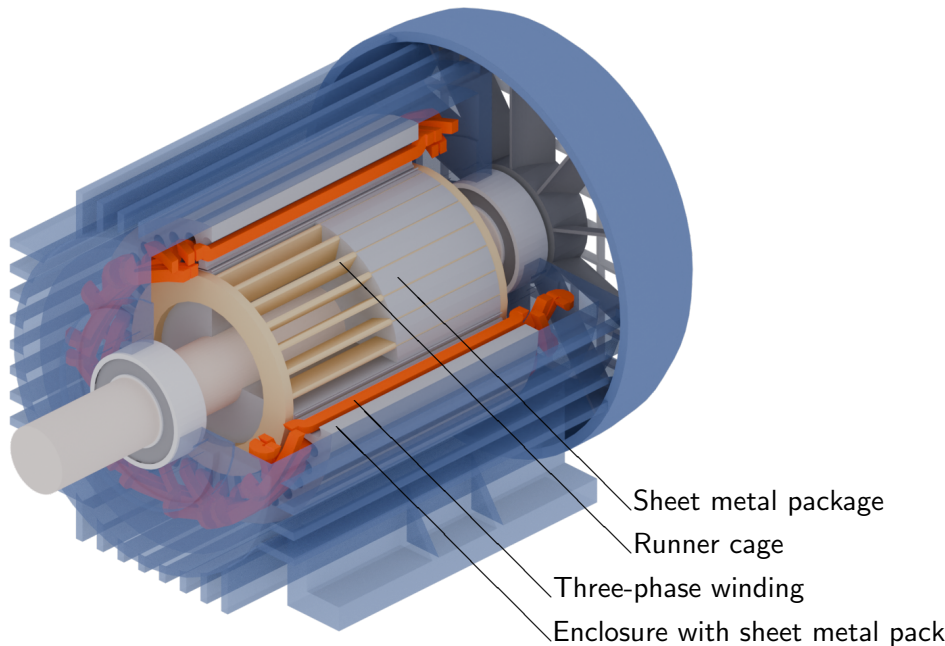


Enclosure with sheet metal pack

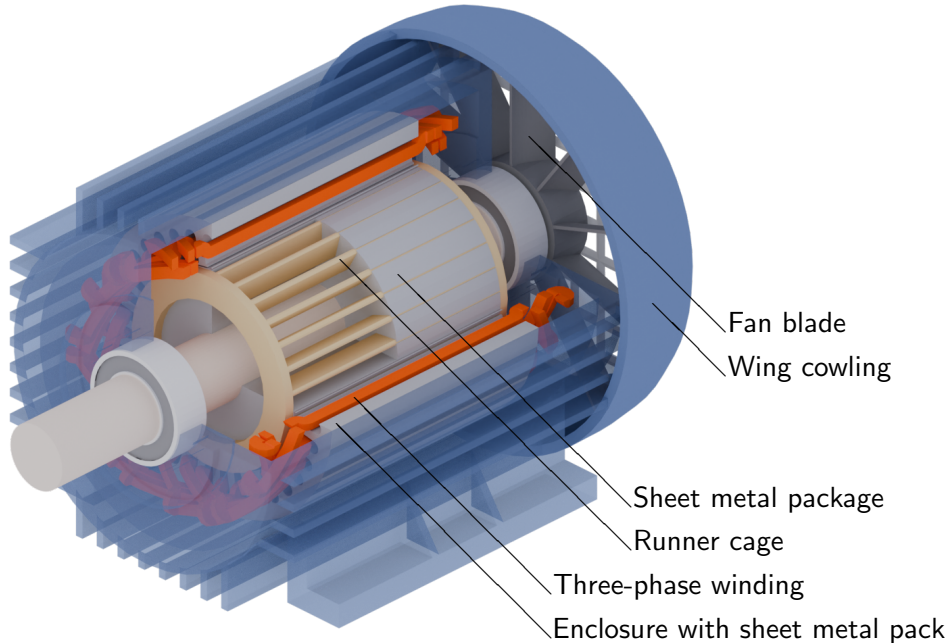
Asynchronous machine design



Asynchronous machine design

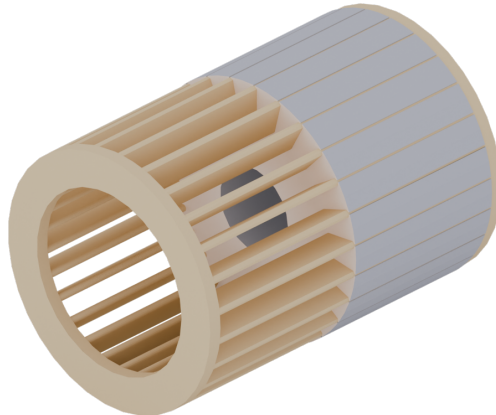


Asynchronous machine design



Cage rotor

- ▶ Conductive rods are located in the grooves of the runner plate package.
- ▶ The rods are connected to each other at the front by short-circuit rings.
- ▶ Together, these components form a cage (similar to a hamster cage).
- ▶ The rods are designed at an angle to reduce harmonics in the rotating magnetic field.



- ▶ The stator generates a rotating magnetic traveling wave in the stationary part with the angular velocity: $\omega_s = \frac{\omega_0}{p}$.

Start-up of the ASM

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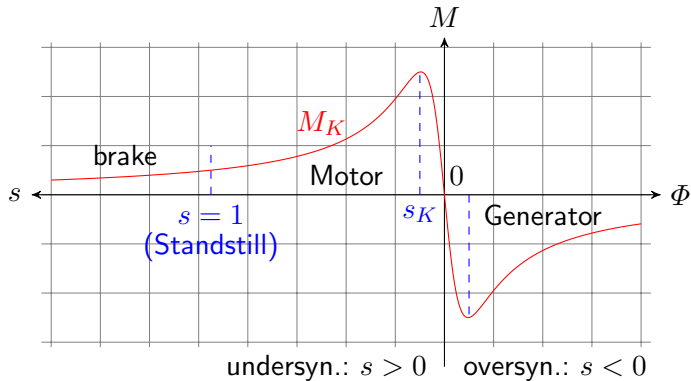
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- ▶ The induced current and the torque of the machine become zero.
- ▶ Due to friction effects, the rotor is slowed down again, leading to the renewed formation of torque.
- ▶ In the steady-state condition, a speed slightly below the synchronous speed is established.

Speed/torque characteristic curve

Slippage:

$$s = \frac{\omega_s - \omega_r}{\omega_s}$$



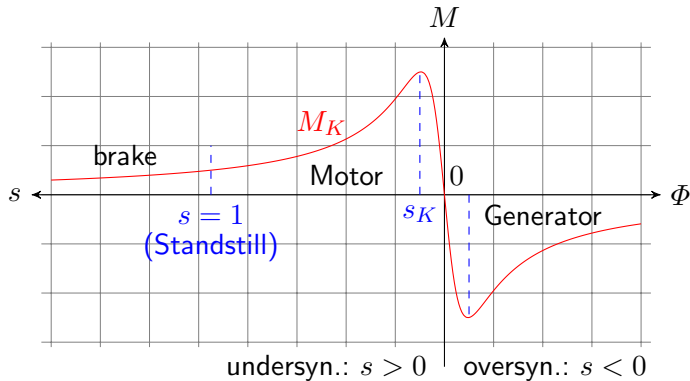
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Kloss's equation

$$M = \frac{2 \cdot M_K}{\frac{s_K}{s} + \frac{s}{s_K}}$$



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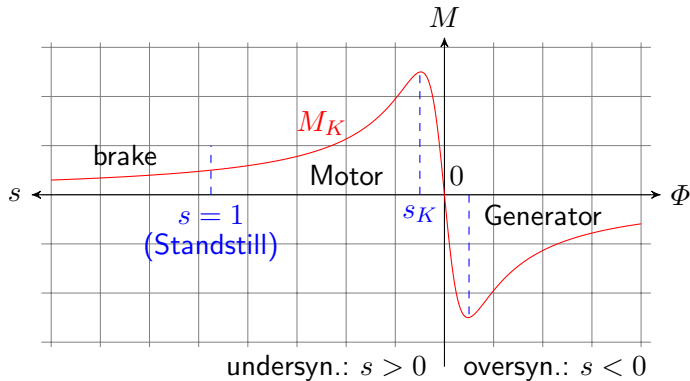
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Slippage corresponds to the percentage losses:

$$\eta \approx 1 - s$$



Example: Asynchronous machine

A three-phase asynchronous motor has the following information on its nameplate:

- ▶ Rated power: 10 kW
- ▶ Rated speed: $1440 \frac{\text{U}}{\text{min}}$
- ▶ Frequency: 50 Hz
- ▶ Breakdown torque: 25%

Calculate the rated torque, the tipping torque and the starting torque.

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$$\begin{aligned} s_N &= \frac{\omega_s - \omega_r}{\omega_s} \\ &= \frac{2\pi \cdot 1500 \frac{1}{60 \text{ s}} - 2\pi \cdot 1440 \frac{1}{60 \text{ s}}}{2\pi \cdot 1500 \frac{1}{60 \text{ s}}} \\ &= 0,04 \end{aligned}$$

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